Data Detective Final Project

Trade Data Validation, Reporting & Reconciliation

Objective: Validate daily trading activity, identify data quality issues, reconcile position changes, and communicate operational discrepancies.

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Data Overview & Workflow

Raw Data Overview

Dataset	Rows	Columns	Key Columns
Trades (sample_trades.csv)	286	13	trade_id, symbol, account, side, quantity, price, trade_date, settlement_date
Positions T0 (positions_t0.csv)	31	3	symbol, account, position_t0
Positions T1 (positions_t1.csv)	30	3	symbol, account, position_t1
Corporate Actions (corp_actions.csv)	9	4	symbol, ex_date, corp_type, corp_shares

- Four datasets were used to analyze trading activity, position changes, and corporate action adjustments.

Data Processing & Reconciliation Workflow

- Raw Data → Data Inspection → Issue Detection → Data Cleaning → Validation → Aggregation → Reporting → Reconciliation → Dashboard

Data Cleaning & Issues Identified

Data Cleaning Results

Issue	Removed	Cleaning Action
Duplicate IDs	8	Keep first occurrence
Zero Quantity	6	Remove rows
Negative quantity	6	
Missing Price	4	
Missing Venue	3	
Invalid Settlement Date	6	
Total	33	

Additional Issues

- Missing positions and corporate action values were handled separately, while price outliers were flagged for review.

Examples of Data Quality Issues

```
1. Duplicate trade IDs:
Duplicate rows: 16
Duplicate IDs: 8
    trade_id symbol account trade_date  side  quantity     price   venue  \
99   T100000   META    ACC2 2023-03-04  SELL      3729  696.7500    BATS
101  T100000   META    ACC2 2023-03-04  SELL      3772    6.9634    BATS
229  T100012  GOOGL    ACC2 2023-03-09   BUY      4116  699.3100    BATS
232  T100012  GOOGL    ACC2 2023-03-09   BUY      4084  705.6100     NaN
187  T100017  GOOGL    ACC1 2023-03-07  SELL      4817  999.3700  NASDAQ
188  T100017  GOOGL    ACC1 2023-03-07  SELL      4836  984.2800  NASDAQ
```

```
2. Zero Quantity Trades
Count: 6
    trade_id symbol account trade_date  side  quantity   price   venue  \
83   T100038   TSLA    ACC1 2023-03-03  SELL         0  452.21  NASDAQ
91   T100047   AMZN    ACC2 2023-03-04  SELL         0  879.13  NASDAQ
153  T100190   AMZN    ACC2 2023-03-06   BUY         0  183.05    NYSE
204  T100039   AMZN    ACC3 2023-03-08   BUY         0  416.40    ARCA
228  T100253  GOOGL    ACC1 2023-03-09   BUY         0   21.21    NYSE
282  T100016   TSLA    ACC3 2023-03-10  SELL         0  596.50    NYSE
```

```
3. Negative quantity trades:
Count: 6
    trade_id symbol account trade_date  side  quantity   price   venue  \
0    T100165   AAPL    ACC1 2023-03-01  SELL     -4851  910.53    ARCA
4    T100248   AMZN    ACC2 2023-03-01   BUY     -4511  308.28  NASDAQ
30   T100069    XOM    ACC1 2023-03-01   BUY      -484  477.96    BATS
242  T100031   NVDA    ACC1 2023-03-09   BUY     -3541  651.66    ARCA
251  T100229    XOM    ACC3 2023-03-09   BUY     -1396  453.45    ARCA
285  T100024    XOM    ACC2 2023-03-10  SELL     -2198  132.98    NYSE
```

Reporting Results & Observations

Generated Data Outputs

- **cleaned_trades.csv**: Cleaned and validated trade data
- **daily_report.csv:** Daily trading activity by symbol
- **symbol_report.csv**: Aggregated trading activity by symbol
- **full_reconciliation.csv**: Position reconciliation results

Key Reporting Insights

- **AAPL**: Highest trading volume (88,307 shares)
- **META**: Largest net buy quantity (+31,813 shares)
- **MSFT**: Largest net sell quantity (−8,691 shares)

```
2. SYMBOL REPORT
  symbol  trade_count  total_quantity  net_qty
0   AAPL           32           88307    11329
2  GOOGL           29           77946      666
4   META           29           75487    31813
6   NVDA           30           68594     -928
9    XOM           23           67511    -7177
5   MSFT           23           63895    -8691
```

Reconciliation Logic

Reconciliation Calculation

- **Actual Change** = Position T1 − Position T0
- **Expected Change** = Net Trade Quantity + Corporate Action Adjustment
- **Reconciliation Difference** = Actual Change − Expected Change

Reconciliation Status

Status	Condition
MATCH	Reconciliation Difference = 0
MISMATCH	Reconciliation Difference ≠ 0
INVALID	Required corporate action data is missing

Important Data Handling

- Missing Positions: Excluded incomplete account-symbol pairs from reconciliation.
- Missing Corporate Action Values: Marked affected results as **INVALID** rather than assuming zero.

Case Study 1: GOOGL

Missing Position & Reconciliation Mismatch

Issue Identified

```
8. Missing position rows:
Count: 1
  symbol account  position_t0  position_t1     _merge
7  GOOGL    ACC2          8742          NaN  left_only
```

- Missing position_t1 for GOOGL in account ACC2.
- The missing record prevented reconciliation for GOOGL ACC2.

Reconciliation Results

Metric	Value
Position T0	14,160
Position T1	9,149
Net Trade Quantity	−1,714
Corporate Action Adjustment	0
Reconciliation Difference	−3,297
Status	**MISMATCH**

Investigation & Handling

- Identified the missing position record.
- Excluded GOOGL ACC2 from reconciliation calculations.
- Recalculated GOOGL using the remaining eligible data.

Key Takeaway

- Excluding incomplete data allowed reconciliation to continue, but a mismatch of 3,297 shares remained.

Case Study 2: JPM

Missing Corporate Action Data & Invalid Reconciliation

Issue Identified

```
10. Missing corporate action shares:
Count: 1
  symbol    ex_date corp_type  corp_shares
8    JPM 2023-03-01       DIV          NaN
```

- JPM had a corporate action record with missing corp_shares.
- The missing value prevented reliable reconciliation.

Reconciliation Results

Metric	Value
Position T0	18,553
Position T1	14,619
Net Trade Quantity	−127
Corporate Action Adjustment	Missing
Reconciliation Difference	N/A
Status	**INVALID**

Investigation & Handling

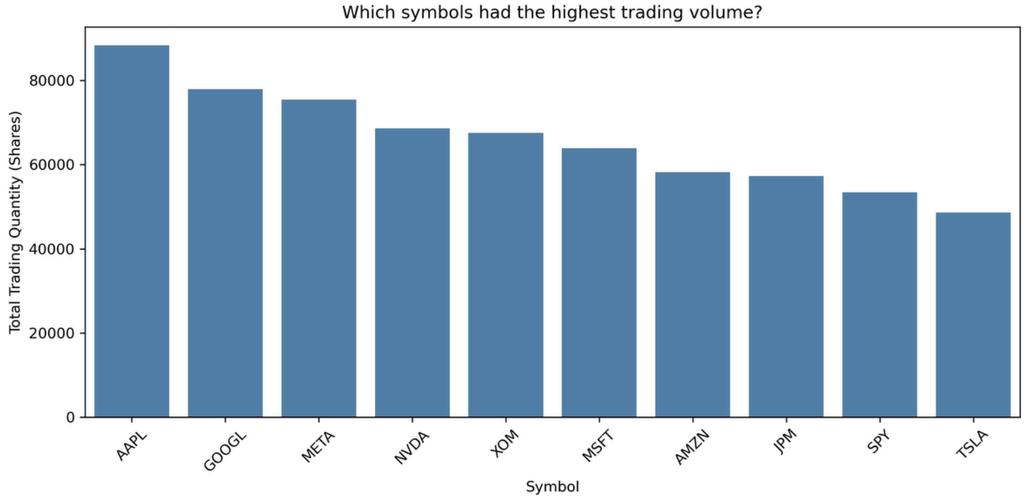
- Identified the missing corporate action value.
- Preserved NaN instead of replacing it with zero.
- Classified the result as INVALID.

Key Takeaway

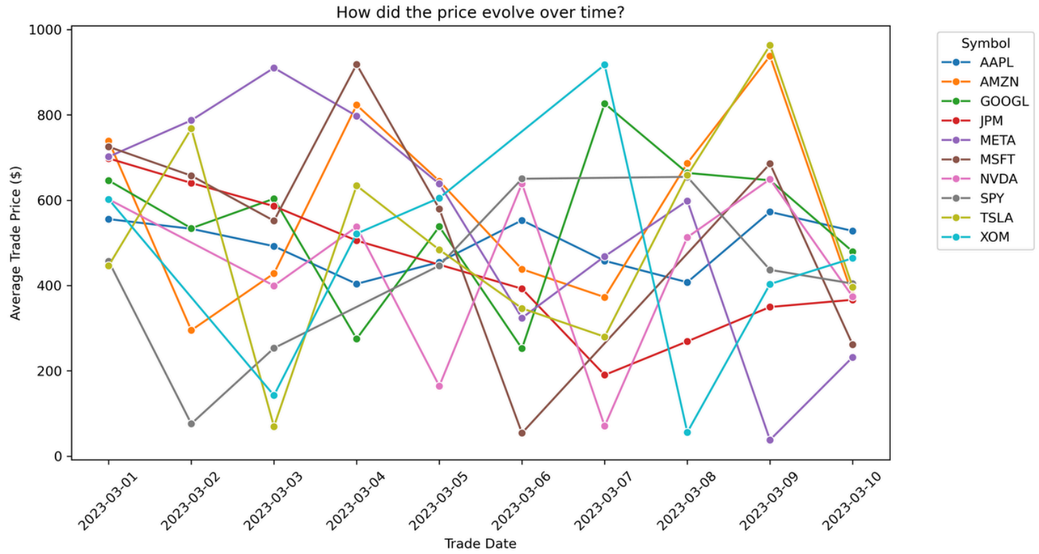
- Missing corporate action values must be flagged to avoid misleading reconciliation results.

Dashboard Highlights & Insights

1. Trading Volume



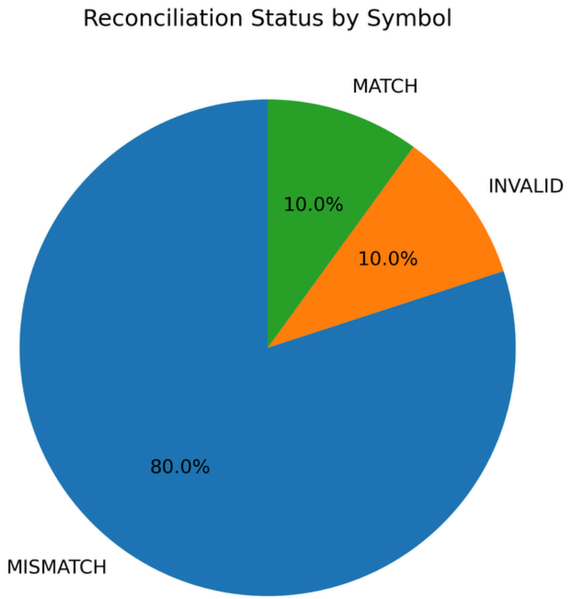
2. Price Trend



- **Trading Volume: AAPL** had the highest trading volume (88,307 shares).
- **Price Trend:** Average trade prices fluctuated over the analyzed period.

Dashboard Highlights & Insights

3. Reconciliation Status



- **Reconciliation Rate:** Only 1 of 10 symbols matched, while 8 had mismatches and 1 was invalid due to missing corporate action data.

Lessons Learned & Limitations

Key Lessons

- Account-level checks helped identify missing positions that symbol-level aggregation could miss.
- Missing data required different handling depending on the issue.
- Some reconciliation differences remained unresolved and required further investigation.

Limitations

- The dataset was simulated using an instructor-provided Python script.
- Corporate action data did not include account information, limiting account-level reconciliation.
- Results were not verified against real trading records.